



Modelling volatility and correlation

Learning Outcomes

In this chapter, you will learn how to

- Discuss the features of data that motivate the use of GARCH models
 - Explain how conditional volatility models are estimated
 - Test for ‘ARCH-effects’ in time series data
 - Produce forecasts from GARCH models
 - Contrast various models from the GARCH family
 - Discuss the three hypothesis testing procedures available under maximum likelihood estimation
 - Construct multivariate conditional volatility models and compare between alternative specifications
 - Estimate univariate and multivariate GARCH models in EViews
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8.1 Motivations: an excursion into non-linearity land

All of the models that have been discussed in chapters 2–7 of this book have been linear in nature – that is, the model is linear in the parameters, so that there is one parameter multiplied by each variable in the model. For example, a structural model could be something like

$$y = \beta_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + u \quad (8.1)$$

or more compactly $y = X\beta + u$. It was additionally assumed that $u_t \sim N(0, \sigma^2)$.

The linear paradigm as described above is a useful one. The properties of linear estimators are very well researched and very well understood. Many models that appear, *prima facie*, to be non-linear, can be made linear

by taking logarithms or some other suitable transformation. However, it is likely that many relationships in finance are intrinsically non-linear. As Campbell, Lo and MacKinlay (1997) state, the payoffs to options are non-linear in some of the input variables, and investors' willingness to trade off returns and risks are also non-linear. These observations provide clear motivations for consideration of non-linear models in a variety of circumstances in order to capture better the relevant features of the data.

Linear structural (and time series) models such as (8.1) are also unable to explain a number of important features common to much financial data, including:

- *Leptokurtosis* – that is, the tendency for financial asset returns to have distributions that exhibit fat tails and excess peakedness at the mean.
- *Volatility clustering or volatility pooling* – the tendency for volatility in financial markets to appear in bunches. Thus large returns (of either sign) are expected to follow large returns, and small returns (of either sign) to follow small returns. A plausible explanation for this phenomenon, which seems to be an almost universal feature of asset return series in finance, is that the information arrivals which drive price changes themselves occur in bunches rather than being evenly spaced over time.
- *Leverage effects* – the tendency for volatility to rise more following a large price fall than following a price rise of the same magnitude.

Campbell, Lo and MacKinlay (1997) broadly define a non-linear data generating process as one where the current value of the series is related non-linearly to current and previous values of the error term

$$y_t = f(u_t, u_{t-1}, u_{t-2}, \dots) \quad (8.2)$$

where u_t is an iid error term and f is a non-linear function. According to Campbell, Lo and MacKinlay, a more workable and slightly more specific definition of a non-linear model is given by the equation

$$y_t = g(u_{t-1}, u_{t-2}, \dots) + u_t \sigma^2(u_{t-1}, u_{t-2}, \dots) \quad (8.3)$$

where g is a function of past error terms only, and σ^2 can be interpreted as a variance term, since it is multiplied by the current value of the error. Campbell, Lo and MacKinlay usefully characterise models with non-linear $g(\bullet)$ as being non-linear in mean, while those with non-linear $\sigma(\bullet)^2$ are characterised as being non-linear in variance.

Models can be linear in mean and variance (e.g. the CLRM, ARMA models) or linear in mean, but non-linear in variance (e.g. GARCH models).

Models could also be classified as non-linear in mean but linear in variance (e.g. bicorrelations models, a simple example of which is of the following form (see Brooks and Hinich, 1999))

$$y_t = \alpha_0 + \alpha_1 y_{t-1} y_{t-2} + u_t \quad (8.4)$$

Finally, models can be non-linear in both mean and variance (e.g. the hybrid threshold model with GARCH errors employed by Brooks, 2001).

8.1.1 *Types of non-linear models*

There are an infinite number of different types of non-linear model. However, only a small number of non-linear models have been found to be useful for modelling financial data. The most popular non-linear financial models are the ARCH or GARCH models used for modelling and forecasting volatility, and switching models, which allow the behaviour of a series to follow different processes at different points in time. Models for volatility and correlation will be discussed in this chapter, with switching models being covered in chapter 9.

8.1.2 *Testing for non-linearity*

How can it be determined whether a non-linear model may potentially be appropriate for the data? The answer to this question should come at least in part from financial theory: a non-linear model should be used where financial theory suggests that the relationship between variables should be such as to require a non-linear model. But the linear versus non-linear choice may also be made partly on statistical grounds – deciding whether a linear specification is sufficient to describe all of the most important features of the data at hand.

So what tools are available to detect non-linear behaviour in financial time series? Unfortunately, ‘traditional’ tools of time series analysis (such as estimates of the autocorrelation or partial autocorrelation function, or ‘spectral analysis’, which involves looking at the data in the frequency domain) are likely to be of little use. Such tools may find no evidence of linear structure in the data, but this would not necessarily imply that the same observations are independent of one another.

However, there are a number of tests for non-linear patterns in time series that are available to the researcher. These tests can broadly be split into two types: general tests and specific tests. General tests, also sometimes called ‘portmanteau’ tests, are usually designed to detect many departures from randomness in data. The implication is that such tests will

detect a variety of non-linear structures in data, although these tests are unlikely to tell the researcher which type of non-linearity is present! Perhaps the simplest general test for non-linearity is Ramsey's RESET test discussed in chapter 4, although there are many other popular tests available. One of the most widely used tests is known as the BDS test (see Brock *et al.*, 1996) named after the three authors who first developed it. BDS is a pure hypothesis test. That is, it has as its null hypothesis that the data are pure noise (completely random), and it has been argued to have power to detect a variety of departures from randomness – linear or non-linear stochastic processes, deterministic chaos, etc. (see Brock *et al.*, 1991). The BDS test follows a standard normal distribution under the null hypothesis. The details of this test, and others, are technical and beyond the scope of this book, although computer code for BDS estimation is now widely available free of charge on the Internet.

As well as applying the BDS test to raw data in an attempt to 'see if there is anything there', another suggested use of the test is as a model diagnostic. The idea is that a proposed model (e.g. a linear model, GARCH, or some other non-linear model) is estimated, and the test applied to the (standardised) residuals in order to 'see what is left'. If the proposed model is adequate, the standardised residuals should be white noise, while if the postulated model is insufficient to capture all of the relevant features of the data, the BDS test statistic for the standardised residuals will be statistically significant. This is an excellent idea in theory, but has difficulties in practice. First, if the postulated model is a non-linear one (such as GARCH), the asymptotic distribution of the test statistic will be altered, so that it will no longer follow a normal distribution. This requires new critical values to be constructed via simulation for every type of non-linear model whose residuals are to be tested. More seriously, if a non-linear model is fitted to the data, any remaining structure is typically garbled, resulting in the test either being unable to detect additional structure present in the data (see Brooks and Henry, 2000) or selecting as adequate a model which is not even in the correct class for that data generating process (see Brooks and Heravi, 1999).

The BDS test is available in EViews. To run it on a given series, simply open the series to be tested (which may be a set of raw data or residuals from an estimated model) so that it appears as a spreadsheet. Then select the **View** menu and **BDS Independence Test....** You will then be offered various options. Further details are given in the *EViews User's Guide*.

Other popular tests for non-linear structure in time series data include the bispectrum test due to Hinich (1982), the bicorrelation test (see Hsieh,

1993; Hinich, 1996; or Brooks and Hinich, 1999 for its multivariate generalisation).

Most applications of the above tests conclude that there is non-linear dependence in financial asset returns series, but that the dependence is best characterised by a GARCH-type process (see Hinich and Patterson, 1985; Baillie and Bollerslev, 1989; Brooks, 1996; and the references therein for applications of non-linearity tests to financial data).

Specific tests, on the other hand, are usually designed to have power to find specific types of non-linear structure. Specific tests are unlikely to detect other forms of non-linearities in the data, but their results will by definition offer a class of models that should be relevant for the data at hand. Examples of specific tests will be offered later in this and subsequent chapters.

8.2 Models for volatility

Modelling and forecasting stock market volatility has been the subject of vast empirical and theoretical investigation over the past decade or so by academics and practitioners alike. There are a number of motivations for this line of inquiry. Arguably, volatility is one of the most important concepts in the whole of finance. Volatility, as measured by the standard deviation or variance of returns, is often used as a crude measure of the total risk of financial assets. Many value-at-risk models for measuring market risk require the estimation or forecast of a volatility parameter. The volatility of stock market prices also enters directly into the Black-Scholes formula for deriving the prices of traded options.

The next few sections will discuss various models that are appropriate to capture the stylised features of volatility, discussed below, that have been observed in the literature.

8.3 Historical volatility

The simplest model for volatility is the historical estimate. Historical volatility simply involves calculating the variance (or standard deviation) of returns in the usual way over some historical period, and this then becomes the volatility forecast for all future periods. The historical average variance (or standard deviation) was traditionally used as the volatility input to options pricing models, although there is a growing body of evidence suggesting that the use of volatility predicted from more

sophisticated time series models will lead to more accurate option valuations (see, for example, Akgiray, 1989; or Chu and Freund, 1996). Historical volatility is still useful as a benchmark for comparing the forecasting ability of more complex time models.

8.4 Implied volatility models

All pricing models for financial options require a volatility estimate or forecast as an input. Given the price of a traded option obtained from transactions data, it is possible to determine the volatility forecast over the lifetime of the option implied by the option's valuation. For example, if the standard Black–Scholes model is used, the option price, the time to maturity, a risk-free rate of interest, the strike price and the current value of the underlying asset, are all either specified in the details of the options contracts or are available from market data. Therefore, given all of these quantities, it is possible to use a numerical procedure, such as the method of bisections or Newton–Raphson to derive the volatility implied by the option (see Watsham and Parramore, 2004). This implied volatility is the market's forecast of the volatility of underlying asset returns over the lifetime of the option.

8.5 Exponentially weighted moving average models

The exponentially weighted moving average (EWMA) is essentially a simple extension of the historical average volatility measure, which allows more recent observations to have a stronger impact on the forecast of volatility than older data points. Under an EWMA specification, the latest observation carries the largest weight, and weights associated with previous observations decline exponentially over time. This approach has two advantages over the simple historical model. First, volatility is in practice likely to be affected more by recent events, which carry more weight, than events further in the past. Second, the effect on volatility of a single given observation declines at an exponential rate as weights attached to recent events fall. On the other hand, the simple historical approach could lead to an abrupt change in volatility once the shock falls out of the measurement sample. And if the shock is still included in a relatively long measurement sample period, then an abnormally large observation will imply that the forecast will remain at an artificially high level even if the market is subsequently tranquil. The exponentially weighted moving

average model can be expressed in several ways, e.g.

$$\sigma_t^2 = (1 - \lambda) \sum_{j=0}^{\infty} \lambda^j (r_{t-j} - \bar{r})^2 \quad (8.5)$$

where σ_t^2 is the estimate of the variance for period t , which also becomes the forecast of future volatility for all periods, $\bar{r}$ is the average return estimated over the observations and λ is the ‘decay factor’, which determines how much weight is given to recent versus older observations. The decay factor could be estimated, but in many studies is set at 0.94 as recommended by RiskMetrics, producers of popular risk measurement software. Note also that RiskMetrics and many academic papers assume that the average return, $\bar{r}$, is zero. For data that is of daily frequency or higher, this is not an unreasonable assumption, and is likely to lead to negligible loss of accuracy since it will typically be very small. Obviously, in practice, an infinite number of observations will not be available on the series, so that the sum in (8.5) must be truncated at some fixed lag. As with exponential smoothing models, the forecast from an EWMA model for all prediction horizons is the most recent weighted average estimate.

It is worth noting two important limitations of EWMA models. First, while there are several methods that could be used to compute the EWMA, the crucial element in each case is to remember that when the infinite sum in (8.5) is replaced with a finite sum of observable data, the weights from the given expression will now sum to less than one. In the case of small samples, this could make a large difference to the computed EWMA and thus a correction may be necessary. Second, most time-series models, such as GARCH (see below), will have forecasts that tend towards the unconditional variance of the series as the prediction horizon increases. This is a good property for a volatility forecasting model to have, since it is well known that volatility series are ‘mean-reverting’. This implies that if they are currently at a high level relative to their historic average, they will have a tendency to fall back towards their average level, while if they are at a low level relative to their historic average, they will have a tendency to rise back towards the average. This feature is accounted for in GARCH volatility forecasting models, but not by EWMA.

8.6 Autoregressive volatility models

Autoregressive volatility models are a relatively simple example from the class of stochastic volatility specifications. The idea is that a time series of observations on some volatility proxy are obtained. The standard

Box–Jenkins-type procedures for estimating autoregressive (or ARMA) models can then be applied to this series. If the quantity of interest in the study is a daily volatility estimate, two natural proxies have been employed in the literature: squared daily returns, or daily range estimators. Producing a series of daily squared returns trivially involves taking a column of observed returns and squaring each observation. The squared return at each point in time, t , then becomes the daily volatility estimate for day t . A range estimator typically involves calculating the log of the ratio of the highest observed price to the lowest observed price for trading day t , which then becomes the volatility estimate for day t

$$\sigma_t^2 = \log \left(\frac{high_t}{low_t} \right) \quad (8.6)$$

Given either the squared daily return or the range estimator, a standard autoregressive model is estimated, with the coefficients β_i estimated using OLS (or maximum likelihood – see below). The forecasts are also produced in the usual fashion discussed in chapter 5 in the context of ARMA models

$$\sigma_t^2 = \beta_0 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2 + \varepsilon_t \quad (8.7)$$

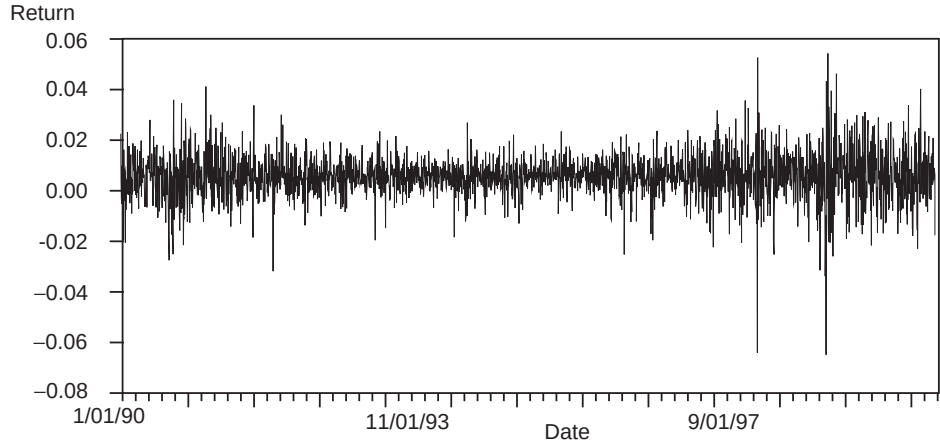
8.7 Autoregressive conditionally heteroscedastic (ARCH) models

One particular non-linear model in widespread usage in finance is known as an ‘ARCH’ model (ARCH stands for ‘autoregressive conditionally heteroscedastic’). To see why this class of models is useful, recall that a typical structural model could be expressed by an equation of the form given in (8.1) above with $u_t \sim N(0, \sigma^2)$. The assumption of the CLRM that the variance of the errors is constant is known as *homoscedasticity* (i.e. it is assumed that $\text{var}(u_t) = \sigma^2$). If the variance of the errors is not constant, this would be known as *heteroscedasticity*. As was explained in chapter 4, if the errors are heteroscedastic, but assumed homoscedastic, an implication would be that standard error estimates could be wrong. It is unlikely in the context of financial time series that the variance of the errors will be constant over time, and hence it makes sense to consider a model that does not assume that the variance is constant, and which describes how the variance of the errors evolves.

Another important feature of many series of financial asset returns that provides a motivation for the ARCH class of models, is known as ‘volatility clustering’ or ‘volatility pooling’. Volatility clustering describes

Figure 8.1

Daily S&P returns
for January
1990–December
1999



the tendency of large changes in asset prices (of either sign) to follow large changes and small changes (of either sign) to follow small changes. In other words, the current level of volatility tends to be positively correlated with its level during the immediately preceding periods. This phenomenon is demonstrated in figure 8.1, which plots daily S&P500 returns for January 1990–December 1999.

The important point to note from figure 8.1 is that *volatility occurs in bursts*. There appears to have been a prolonged period of relative tranquility in the market during the mid-1990s, evidenced by only relatively small positive and negative returns. On the other hand, during mid-1997 to late 1998, there was far more volatility, when many large positive and large negative returns were observed during a short space of time. Abusing the terminology slightly, it could be stated that ‘volatility is autocorrelated’.

How could this phenomenon, which is common to many series of financial asset returns, be parameterised (modelled)? One approach is to use an ARCH model. To understand how the model works, a definition of the conditional variance of a random variable, u_t , is required. The distinction between the conditional and unconditional variances of a random variable is exactly the same as that of the conditional and unconditional mean. The conditional variance of u_t may be denoted σ_t^2 , which is written as

$$\sigma_t^2 = \text{var}(u_t | u_{t-1}, u_{t-2}, \dots) = E[(u_t - E(u_t))^2 | u_{t-1}, u_{t-2}, \dots] \quad (8.8)$$

It is usually assumed that $E(u_t) = 0$, so

$$\sigma_t^2 = \text{var}(u_t | u_{t-1}, u_{t-2}, \dots) = E[u_t^2 | u_{t-1}, u_{t-2}, \dots] \quad (8.9)$$

Equation (8.9) states that the conditional variance of a zero mean normally distributed random variable u_t is equal to the conditional expected

value of the square of u_t . Under the ARCH model, the ‘autocorrelation in volatility’ is modelled by allowing the conditional variance of the error term, σ_t^2 , to depend on the immediately previous value of the squared error

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 \quad (8.10)$$

The above model is known as an ARCH(1), since the conditional variance depends on only one lagged squared error. Notice that (8.10) is only a partial model, since nothing has been said yet about the conditional mean. Under ARCH, the conditional mean equation (which describes how the dependent variable, y_t , varies over time) could take almost any form that the researcher wishes. One example of a full model would be

$$y_t = \beta_1 + \beta_2 x_{2t} + \beta_3 x_{3t} + \beta_4 x_{4t} + u_t \quad u_t \sim N(0, \sigma_t^2) \quad (8.11)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 \quad (8.12)$$

The model given by (8.11) and (8.12) could easily be extended to the general case where the error variance depends on q lags of squared errors, which would be known as an ARCH(q) model:

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \alpha_2 u_{t-2}^2 + \cdots + \alpha_q u_{t-q}^2 \quad (8.13)$$

Instead of calling the conditional variance σ_t^2 , in the literature it is often called h_t , so that the model would be written

$$y_t = \beta_1 + \beta_2 x_{2t} + \beta_3 x_{3t} + \beta_4 x_{4t} + u_t \quad u_t \sim N(0, h_t) \quad (8.14)$$

$$h_t = \alpha_0 + \alpha_1 u_{t-1}^2 + \alpha_2 u_{t-2}^2 + \cdots + \alpha_q u_{t-q}^2 \quad (8.15)$$

The remainder of this chapter will use σ_t^2 to denote the conditional variance at time t , except for computer instructions where h_t will be used since it is easier not to use Greek letters.

8.7.1 Another way of expressing ARCH models

For illustration, consider an ARCH(1). The model can be expressed in two ways that look different but are in fact identical. The first is as given in (8.11) and (8.12) above. The second way would be as follows

$$y_t = \beta_1 + \beta_2 x_{2t} + \beta_3 x_{3t} + \beta_4 x_{4t} + u_t \quad (8.16)$$

$$u_t = v_t \sigma_t \quad v_t \sim N(0, 1) \quad (8.17)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 \quad (8.18)$$

The form of the model given in (8.11) and (8.12) is more commonly presented, although specifying the model as in (8.16)–(8.18) is required in

order to use a GARCH process in a simulation study (see chapter 12). To show that the two methods for expressing the model are equivalent, consider that in (8.17), v_t is normally distributed with zero mean and unit variance, so that u_t will also be normally distributed with zero mean and variance σ_t^2 .

8.7.2 Non-negativity constraints

Since h_t is a conditional variance, its value must always be strictly positive; a negative variance at any point in time would be meaningless. The variables on the RHS of the conditional variance equation are all squares of lagged errors, and so by definition will not be negative. In order to ensure that these always result in positive conditional variance estimates, all of the coefficients in the conditional variance are usually required to be non-negative. If one or more of the coefficients were to take on a negative value, then for a sufficiently large lagged squared innovation term attached to that coefficient, the fitted value from the model for the conditional variance could be negative. This would clearly be nonsensical. So, for example, in the case of (8.18), the non-negativity condition would be $\alpha_0 \geq 0$ and $\alpha_1 \geq 0$. More generally, for an ARCH(q) model, all coefficients would be required to be non-negative: $\alpha_i \geq 0 \forall i = 0, 1, 2, \dots, q$. In fact, this is a sufficient but not necessary condition for non-negativity of the conditional variance (i.e. it is a slightly stronger condition than is actually necessary).

8.7.3 Testing for 'ARCH effects'

A test for determining whether 'ARCH-effects' are present in the residuals of an estimated model may be conducted using the steps outlined in box 8.1.

Thus, the test is one of a joint null hypothesis that all q lags of the squared residuals have coefficient values that are not significantly different from zero. If the value of the test statistic is greater than the critical value from the χ^2 distribution, then reject the null hypothesis. The test can also be thought of as a test for autocorrelation in the squared residuals. As well as testing the residuals of an estimated model, the ARCH test is frequently applied to raw returns data.

8.7.4 Testing for 'ARCH effects' in exchange rate returns using EViews

Before estimating a GARCH-type model, it is sensible first to compute the Engle (1982) test for ARCH effects to make sure that this class of models is appropriate for the data. This exercise (and the remaining exercises of this chapter), will employ returns on the daily exchange rates where there are

Box 8.1 Testing for 'ARCH effects'

- (1) Run any postulated linear regression of the form given in the equation above, e.g.

$$y_t = \beta_1 + \beta_2 x_{2t} + \beta_3 x_{3t} + \beta_4 x_{4t} + u_t \quad (8.19)$$

saving the residuals, $\hat{u}_t$.

- (2) Square the residuals, and regress them on q own lags to test for ARCH of order q , i.e. run the regression

$$\hat{u}_t^2 = \gamma_0 + \gamma_1 \hat{u}_{t-1}^2 + \gamma_2 \hat{u}_{t-2}^2 + \cdots + \gamma_q \hat{u}_{t-q}^2 + v_t \quad (8.20)$$

where v_t is an error term.

Obtain R^2 from this regression.

- (3) The test statistic is defined as TR^2 (the number of observations multiplied by the coefficient of multiple correlation) from the last regression, and is distributed as a $\chi^2(q)$.
- (4) The null and alternative hypotheses are

$$H_0 : \gamma_1 = 0 \text{ and } \gamma_2 = 0 \text{ and } \gamma_3 = 0 \text{ and } \dots \text{ and } \gamma_q = 0$$

$$H_1 : \gamma_1 \neq 0 \text{ or } \gamma_2 \neq 0 \text{ or } \gamma_3 \neq 0 \text{ or } \dots \text{ or } \gamma_q \neq 0$$

1,827 observations. Models of this kind are inevitably more data intensive than those based on simple linear regressions, and hence, everything else being equal, they work better when the data are sampled daily rather than at a lower frequency.

A test for the presence of ARCH in the residuals is calculated by regressing the squared residuals on a constant and p lags, where p is set by the user. As an example, assume that p is set to 5. The first step is to estimate a linear model so that the residuals can be tested for ARCH. From the main menu, select **Quick** and then select **Estimate Equation**. In the Equation Specification Editor, input **rgbp c ar(1) ma(1)** which will estimate an ARMA(1,1) for the pound-dollar returns.¹ Select the **Least Squares (NLA and ARMA)** procedure to estimate the model, using the whole sample period and press the **OK** button (output not shown).

The next step is to click on **View** from the Equation Window and to select **Residual Tests** and then **Heteroskedasticity Tests....** In the 'Test type' box, choose **ARCH** and the number of lags to include is 5, and press **OK**. The output below shows the Engle test results.

¹ Note that the (1,1) order has been chosen entirely arbitrarily at this stage. However, it is important to give some thought to the type and order of model used even if it is not of direct interest in the problem at hand (which will later be termed the 'conditional mean' equation), since the variance is measured around the mean and therefore any mis-specification in the mean is likely to lead to a mis-specified variance.

Heteroskedasticity Test: ARCH

F-statistic	5.909063	Prob. F(5,1814)	0.0000
Obs*R-squared	29.16797	Prob. Chi-Square(5)	0.0000

Test Equation:

Dependent Variable: RESID^2

Method: Least Squares

Date: 09/06/07 Time: 14:41

Sample (adjusted): 7/14/2002 7/07/2007

Included observations: 1820 after adjustments

	Coefficient	Std. Error	t-Statistic	Prob.
C	0.154689	0.011369	13.60633	0.0000
RESID^2(-1)	0.118068	0.023475	5.029627	0.0000
RESID^2(-2)	-0.006579	0.023625	-0.278463	0.7807
RESID^2(-3)	0.029000	0.023617	1.227920	0.2196
RESID^2(-4)	-0.032744	0.023623	-1.386086	0.1659
RESID^2(-5)	-0.020316	0.023438	-0.866798	0.3862
R-squared	0.016026	Mean dependent var		0.169496
Adjusted R-squared	0.013314	S.D. dependent var		0.344448
S.E. of regression	0.342147	Akaike info criterion		0.696140
Sum squared resid	212.3554	Schwarz criterion		0.714293
Log likelihood	-627.4872	Hannan-Quinn criter.		0.702837
F-statistic	5.909063	Durbin-Watson stat		1.995904
Prob(F-statistic)	0.000020			

Both the F -version and the LM -statistic are very significant, suggesting the presence of ARCH in the pound-dollar returns.

8.7.5 Limitations of ARCH(q) models

ARCH provided a framework for the analysis and development of time series models of volatility. However, ARCH models themselves have rarely been used in the last decade or more, since they bring with them a number of difficulties:

- How should the value of q , the number of lags of the squared residual in the model, be decided? One approach to this problem would be the use of a likelihood ratio test, discussed later in this chapter, although there is no clearly best approach.
- The value of q , the number of lags of the squared error that are required to capture all of the dependence in the conditional variance, might be very large. This would result in a large conditional variance model that was not parsimonious. Engle (1982) circumvented this problem by

specifying an arbitrary linearly declining lag length on an ARCH(4)

$$\sigma_t^2 = \gamma_0 + \gamma_1(0.4\hat{u}_{t-1}^2 + 0.3\hat{u}_{t-2}^2 + 0.2\hat{u}_{t-3}^2 + 0.1\hat{u}_{t-4}^2) \quad (8.21)$$

such that only two parameters are required in the conditional variance equation (γ_0 and γ_1), rather than the five which would be required for an unrestricted ARCH(4).

- *Non-negativity constraints might be violated.* Everything else equal, the more parameters there are in the conditional variance equation, the more likely it is that one or more of them will have negative estimated values.

A natural extension of an ARCH(q) model which overcomes some of these problems is a GARCH model. In contrast with ARCH, GARCH models are extremely widely employed in practice.

8.8 Generalised ARCH (GARCH) models

The GARCH model was developed independently by Bollerslev (1986) and Taylor (1986). The GARCH model allows the conditional variance to be dependent upon previous own lags, so that the conditional variance equation in the simplest case is now

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (8.22)$$

This is a GARCH(1,1) model. σ_t^2 is known as the *conditional variance* since it is a one-period ahead estimate for the variance calculated based on any past information thought relevant. Using the GARCH model it is possible to interpret the current fitted variance, h_t , as a weighted function of a long-term average value (dependent on α_0), information about volatility during the previous period ($\alpha_1 u_{t-1}^2$) and the fitted variance from the model during the previous period ($\beta \sigma_{t-1}^2$). Note that the GARCH model can be expressed in a form that shows that it is effectively an ARMA model for the conditional variance. To see this, consider that the squared return at time t relative to the conditional variance is given by

$$\varepsilon_t = u_t^2 - \sigma_t^2 \quad (8.23)$$

or

$$\sigma_t^2 = u_t^2 - \varepsilon_t \quad (8.24)$$

Using the latter expression to substitute in for the conditional variance in (8.22)

$$u_t^2 - \varepsilon_t = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta (u_{t-1}^2 - \varepsilon_{t-1}) \quad (8.25)$$

Rearranging

$$u_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta u_{t-1}^2 - \beta \varepsilon_{t-1} + \varepsilon_t \quad (8.26)$$

so that

$$u_t^2 = \alpha_0 + (\alpha_1 + \beta) u_{t-1}^2 - \beta \varepsilon_{t-1} + \varepsilon_t \quad (8.27)$$

This final expression is an ARMA(1,1) process for the squared errors.

Why is GARCH a better and therefore a far more widely used model than ARCH? The answer is that the former is more parsimonious, and avoids overfitting. Consequently, the model is less likely to breach non-negativity constraints. In order to illustrate why the model is parsimonious, first take the conditional variance equation in the GARCH(1,1) case, subtract 1 from each of the time subscripts of the conditional variance equation in (8.22), so that the following expression would be obtained

$$\sigma_{t-1}^2 = \alpha_0 + \alpha_1 u_{t-2}^2 + \beta \sigma_{t-2}^2 \quad (8.28)$$

and subtracting 1 from each of the time subscripts again

$$\sigma_{t-2}^2 = \alpha_0 + \alpha_1 u_{t-3}^2 + \beta \sigma_{t-3}^2 \quad (8.29)$$

Substituting into (8.22) for σ_{t-1}^2

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta (\alpha_0 + \alpha_1 u_{t-2}^2 + \beta \sigma_{t-2}^2) \quad (8.30)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \alpha_0 \beta + \alpha_1 \beta u_{t-2}^2 + \beta^2 \sigma_{t-2}^2 \quad (8.31)$$

Now substituting into (8.31) for σ_{t-2}^2

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \alpha_0 \beta + \alpha_1 \beta u_{t-2}^2 + \beta^2 (\alpha_0 + \alpha_1 u_{t-3}^2 + \beta \sigma_{t-3}^2) \quad (8.32)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \alpha_0 \beta + \alpha_1 \beta u_{t-2}^2 + \alpha_0 \beta^2 + \alpha_1 \beta^2 u_{t-3}^2 + \beta^3 \sigma_{t-3}^2 \quad (8.33)$$

$$\sigma_t^2 = \alpha_0 (1 + \beta + \beta^2) + \alpha_1 u_{t-1}^2 (1 + \beta L + \beta^2 L^2) + \beta^3 \sigma_{t-3}^2 \quad (8.34)$$

An infinite number of successive substitutions of this kind would yield

$$\sigma_t^2 = \alpha_0 (1 + \beta + \beta^2 + \dots) + \alpha_1 u_{t-1}^2 (1 + \beta L + \beta^2 L^2 + \dots) + \beta^\infty \sigma_0^2 \quad (8.35)$$

The first expression on the RHS of (8.35) is simply a constant, and as the number of observations tends to infinity, β^∞ will tend to zero. Hence, the GARCH(1,1) model can be written as

$$\sigma_t^2 = \gamma_0 + \alpha_1 u_{t-1}^2 (1 + \beta L + \beta^2 L^2 + \dots) \quad (8.36)$$

$$= \gamma_0 + \gamma_1 u_{t-1}^2 + \gamma_2 u_{t-2}^2 + \dots, \quad (8.37)$$

which is a restricted infinite order ARCH model. Thus the GARCH(1,1) model, containing only three parameters in the conditional variance

equation, is a very parsimonious model, that allows an infinite number of past squared errors to influence the current conditional variance.

The GARCH(1,1) model can be extended to a GARCH(p, q) formulation, where the current conditional variance is parameterised to depend upon q lags of the squared error and p lags of the conditional variance

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \alpha_2 u_{t-2}^2 + \cdots + \alpha_q u_{t-q}^2 + \beta_1 \sigma_{t-1}^2 + \beta_2 \sigma_{t-2}^2 + \cdots + \beta_p \sigma_{t-p}^2 \quad (8.38)$$

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i u_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2 \quad (8.39)$$

But in general a GARCH(1,1) model will be sufficient to capture the volatility clustering in the data, and rarely is any higher order model estimated or even entertained in the academic finance literature.

8.8.1 The unconditional variance under a GARCH specification

The conditional variance is changing, but the unconditional variance of u_t is constant and given by

$$\text{var}(u_t) = \frac{\alpha_0}{1 - (\alpha_1 + \beta)} \quad (8.40)$$

so long as $\alpha_1 + \beta < 1$. For $\alpha_1 + \beta \geq 1$, the unconditional variance of u_t is not defined, and this would be termed ‘non-stationarity in variance’. $\alpha_1 + \beta = 1$ would be known as a ‘unit root in variance’, also termed ‘Integrated GARCH’ or IGARCH. Non-stationarity in variance does not have a strong theoretical motivation for its existence, as would be the case for non-stationarity in the mean (e.g. of a price series). Furthermore, a GARCH model whose coefficients imply non-stationarity in variance would have some highly undesirable properties. One illustration of these relates to the forecasts of variance made from such models. For stationary GARCH models, conditional variance forecasts converge upon the long-term average value of the variance as the prediction horizon increases (see below). For IGARCH processes, this convergence will not happen, while for $\alpha_1 + \beta > 1$, the conditional variance forecast will tend to infinity as the forecast horizon increases!

8.9 Estimation of ARCH/GARCH models

Since the model is no longer of the usual linear form, OLS cannot be used for GARCH model estimation. There are a variety of reasons for this, but the simplest and most fundamental is that OLS minimises the residual

Box 8.2 Estimating an ARCH or GARCH model

- (1) Specify the appropriate equations for the mean and the variance – e.g. an AR(1)-GARCH(1,1) model

$$y_t = \mu + \phi y_{t-1} + u_t, u_t \sim N(0, \sigma_t^2) \quad (8.41)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (8.42)$$

- (2) Specify the log-likelihood function (*LLF*) to maximise under a normality assumption for the disturbances

$$L = -\frac{T}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^T \log(\sigma_t^2) - \frac{1}{2} \sum_{t=1}^T (y_t - \mu - \phi y_{t-1})^2 / \sigma_t^2 \quad (8.43)$$

- (3) The computer will maximise the function and generate parameter values that maximise the *LLF* and will construct their standard errors.

sum of squares. The RSS depends only on the parameters in the conditional mean equation, and not the conditional variance, and hence RSS minimisation is no longer an appropriate objective.

In order to estimate models from the GARCH family, another technique known as *maximum likelihood* is employed. Essentially, the method works by finding the most likely values of the parameters given the actual data. More specifically, a log-likelihood function is formed and the values of the parameters that maximise it are sought. Maximum likelihood estimation can be employed to find parameter values for both linear and non-linear models. The steps involved in actually estimating an ARCH or GARCH model are shown in box 8.2.

The following section will elaborate on points 2 and 3 above, explaining how the *LLF* is derived.

8.9.1 Parameter estimation using maximum likelihood

As stated above, under maximum likelihood estimation, a set of parameter values are chosen that are most likely to have produced the observed data. This is done by first forming a *likelihood function*, denoted *LF*. *LF* will be a multiplicative function of the actual data, which will consequently be difficult to maximise with respect to the parameters. Therefore, its logarithm is taken in order to turn *LF* into an additive function of the sample data, i.e. the *LLF*. A derivation of the maximum likelihood (ML) estimator in the context of the simple bivariate regression model with homoscedasticity is given in the appendix to this chapter. Essentially, deriving the ML estimators involves differentiating the *LLF* with respect to the parameters. But how does this help in estimating heteroscedastic models? How can the

method outlined in the appendix for homoscedastic models be modified for application to GARCH model estimation?

In the context of conditional heteroscedasticity models, the model is $y_t = \mu + \phi y_{t-1} + u_t$, $u_t \sim N(0, \sigma_t^2)$, so that the variance of the errors has been modified from being assumed constant, σ^2 , to being time-varying, σ_t^2 , with the equation for the conditional variance as previously. The *LLF* relevant for a GARCH model can be constructed in the same way as for the homoscedastic case by replacing

$$\frac{T}{2} \log \sigma^2$$

with the equivalent for time-varying variance

$$\frac{1}{2} \sum_{t=1}^T \log \sigma_t^2$$

and replacing σ^2 in the denominator of the last part of the expression with σ_t^2 (see the appendix to this chapter). Derivation of this result from first principles is beyond the scope of this text, but the log-likelihood function for the above model with time-varying conditional variance and normally distributed errors is given by (8.43) in box 8.2.

Intuitively, maximising the *LLF* involves jointly minimising

$$\sum_{t=1}^T \log \sigma_t^2$$

and

$$\sum_{t=1}^T \frac{(y_t - \mu - \phi y_{t-1})^2}{\sigma_t^2}$$

(since these terms appear preceded with a negative sign in the *LLF*, and

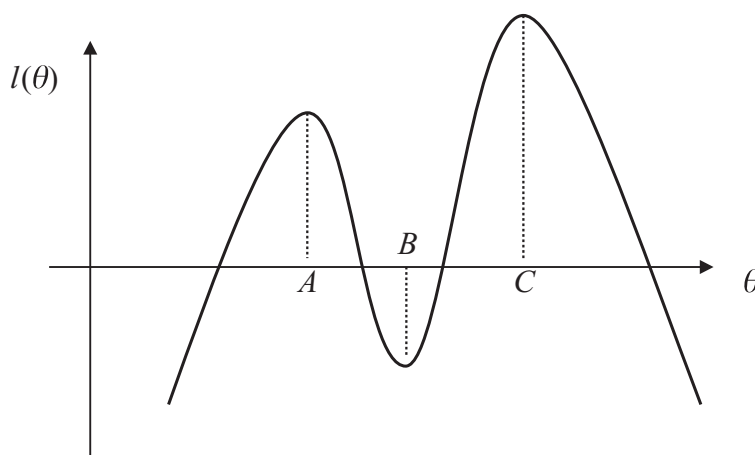
$$-\frac{T}{2} \log(2\pi)$$

is just a constant with respect to the parameters). Minimising these terms jointly also implies minimising the error variance, as described in chapter 3. Unfortunately, maximising the *LLF* for a model with time-varying variances is trickier than in the homoscedastic case. Analytical derivatives of the *LLF* in (8.43) with respect to the parameters have been developed, but only in the context of the simplest examples of GARCH specifications. Moreover, the resulting formulae are complex, so a numerical procedure is often used instead to maximise the log-likelihood function.

Essentially, all methods work by ‘searching’ over the parameter-space until the values of the parameters that maximise the log-likelihood

Figure 8.2

The problem of local optima in maximum likelihood estimation



function are found. EViews employs an iterative technique for maximising the *LLF*. This means that, given a set of initial guesses for the parameter estimates, these parameter values are updated at each iteration until the program determines that an optimum has been reached. If the *LLF* has only one maximum with respect to the parameter values, any optimisation method should be able to find it – although some methods will take longer than others. A detailed presentation of the various methods available is beyond the scope of this book. However, as is often the case with non-linear models such as GARCH, the *LLF* can have many local maxima, so that different algorithms could find different local maxima of the *LLF*. Hence readers should be warned that different optimisation procedures could lead to different coefficient estimates and especially different estimates of the standard errors (see Brooks, Burke and Persaud, 2001 or 2003 for details). In such instances, a good set of initial parameter guesses is essential.

Local optima or multimodalities in the likelihood surface present potentially serious drawbacks with the maximum likelihood approach to estimating the parameters of a GARCH model, as shown in figure 8.2.

Suppose that the model contains only one parameter, θ , so that the log-likelihood function is to be maximised with respect to this one parameter. In figure 8.2, the value of the *LLF* for each value of θ is denoted $l(\theta)$. Clearly, $l(\theta)$ reaches a global maximum when $\theta = C$, and a local maximum when $\theta = A$. This demonstrates the importance of good initial guesses for the parameters. Any initial guesses to the left of B are likely to lead to the selection of A rather than C . The situation is likely to be even worse in practice, since the log-likelihood function will be maximised with respect to several parameters, rather than one, and there could be

Box 8.3 Using maximum likelihood estimation in practice

- (1) Set up the *LLF*.
- (2) Use regression to get *initial estimates* for the mean parameters.
- (3) Choose some initial guesses for the *conditional variance parameters*. In most software packages, the default initial values for the conditional variance parameters would be zero. This is unfortunate since zero parameter values often yield a local maximum of the likelihood function. So if possible, set plausible initial values away from zero.
- (4) Specify a *convergence criterion* – either by criterion or by value. When ‘by criterion’ is selected, the package will continue to search for ‘better’ parameter values that give a higher value of the *LLF* until the change in the value of the *LLF* between iterations is less than the specified convergence criterion. Choosing ‘by value’ will lead to the software searching until the change in the coefficient estimates are small enough. The default convergence criterion for EViews is 0.001, which means that convergence is achieved and the program will stop searching if the biggest percentage change in any of the coefficient estimates for the most recent iteration is smaller than 0.1%.

many local optima. Another possibility that would make optimisation difficult is when the *LLF* is flat around the maximum. So, for example, if the peak corresponding to C in figure 8.2, were flat rather than sharp, a range of values for θ could lead to very similar values for the *LLF*, making it difficult to choose between them.

So, to explain again in more detail, the optimisation is done in the way shown in box 8.3.

The optimisation methods employed by EViews are based on the determination of the first and second derivatives of the log-likelihood function with respect to the parameter values at each iteration, known as the gradient and Hessian (the matrix of second derivatives of the *LLF* w.r.t the parameters), respectively. An algorithm for optimisation due to Berndt, Hall, Hall and Hausman (1974), known as BHHH, is available in EViews. BHHH employs only first derivatives (calculated numerically rather than analytically) and approximations to the second derivatives are calculated. Not calculating the actual Hessian at each iteration at each time step increases computational speed, but the approximation may be poor when the *LLF* is a long way from its maximum value, requiring more iterations to reach the optimum. The Marquardt algorithm, available in EViews, is a modification of BHHH (both of which are variants on the Gauss–Newton method) that incorporates a ‘correction’, the effect of which is to push the coefficient estimates more quickly to their optimal values. All of these optimisation methods are described in detail in Press *et al.* (1992).

8.9.2 Non-normality and maximum likelihood

Recall that the conditional normality assumption for u_t is essential in specifying the likelihood function. It is possible to test for non-normality using the following representation

$$u_t = v_t \sigma_t, v_t \sim N(0, 1) \quad (8.44)$$

$$\sigma_t = \sqrt{\alpha_0 + \alpha_1 u_{t-1}^2 + \beta \sigma_{t-1}^2} \quad (8.45)$$

Note that one would not expect u_t to be normally distributed – it is a $N(0, \sigma_t^2)$ disturbance term from the regression model, which will imply it is likely to have fat tails. A plausible method to test for normality would be to construct the statistic

$$v_t = \frac{u_t}{\sigma_t} \quad (8.46)$$

which would be the model disturbance at each point in time t divided by the conditional standard deviation at that point in time. Thus, it is the v_t that are assumed to be normally distributed, not u_t . The sample counterpart would be

$$\hat{v}_t = \frac{\hat{u}_t}{\hat{\sigma}_t} \quad (8.47)$$

which is known as a standardised residual. Whether the $\hat{v}_t$ are normal can be examined using any standard normality test, such as the Bera–Jarque. Typically, $\hat{v}_t$ are still found to be leptokurtic, although less so than the $\hat{u}_t$. The upshot is that the GARCH model is able to capture some, although not all, of the leptokurtosis in the unconditional distribution of asset returns.

Is it a problem if $\hat{v}_t$ are not normally distributed? Well, the answer is ‘not really’. Even if the conditional normality assumption does not hold, the parameter estimates will still be consistent if the equations for the mean and variance are correctly specified. However, in the context of non-normality, the usual standard error estimates will be inappropriate, and a different variance–covariance matrix estimator that is robust to non-normality, due to Bollerslev and Wooldridge (1992), should be used. This procedure (i.e. maximum likelihood with Bollerslev–Wooldridge standard errors) is known as *quasi-maximum likelihood*, or QML.

8.9.3 Estimating GARCH models in EViews

To estimate a GARCH-type model, open the equation specification dialog by selecting **Quick/Estimate Equation** or by selecting **Object/New Object/Equation...** Select ARCH from the ‘Estimation Settings’ selection box. The window in screenshot 8.1 will open.

Screenshot 8.1

Estimating a
GARCH-type model

Equation Estimation

Specification Options

Mean equation
Dependent followed by regressors and ARMA terms OR explicit
ARCH-M: None

Variance and distribution specification
Model: GARCH/TARCH
Order:
ARCH: 1 Threshold order: 0
GARCH: 1
Restrictions: None
Variance regressors:
Error distribution: Normal (Gaussian)

Estimation settings
Method: ARCH - Autoregressive Conditional Heteroskedasticity
Sample: 7/07/2002 7/07/2007

OK Cancel

It is necessary to specify both the mean and the variance equations, as well as the estimation technique and sample.

The mean equation

The specification of the mean equation should be entered in the dependent variable edit box. Enter the specification by listing the dependent variable followed by the regressors. The constant term 'C' should also be included. If your specification includes an ARCH-M term (see later in this chapter), you should click on the appropriate button in the upper RHS of the dialog box to select the conditional standard deviation, the conditional variance, or the log of the conditional variance.

The variance equation

The edit box labelled 'Variance regressors' is where variables that are to be included in the variance specification should be listed. Note that EViews will always include a constant in the conditional variance, so that it is not necessary to add 'C' to the variance regressor list. Similarly, it is not

necessary to include the ARCH or GARCH terms in this box as they will be dealt with in other parts of the dialog box. Instead, enter here any exogenous variables or dummies that you wish to include in the conditional variance equation, or (as is usually the case), just leave this box blank.

Variance and distribution specification

Under the 'Variance and distribution Specification' label, choose the number of ARCH and GARCH terms. The default is to estimate with one ARCH and one GARCH term (i.e. one lag of the squared errors and one lag of the conditional variance, respectively). To estimate the standard GARCH model, leave the default 'GARCH/TARCH'. The other entries in this box describe more complicated variants of the standard GARCH specification, which are described in later sections of this chapter.

Estimation options

EViews provides a number of optional estimation settings. Clicking on the Options tab gives the options in screenshot 8.2 to be filled out as required.

Screenshot 8.2

GARCH model
estimation options

Equation Estimation

Specification **Options**

Backcasting

☒ Backcast presample MA terms

Presample variance:

Backcast with parameter = 0.7

Coefficient covariance

☐ Heteroskedasticity consistent covariance (Bollerslev-Wooldridge)

Derivatives

Select method to favor:

☒ Accuracy

☐ Speed

☐ Use numeric only

Iterative process

Max Iterations: 500

Convergence: 0.0001

Starting coefficient values:

OLS/TSLS

☐ Display settings

Optimization algorithm

☒ Marquardt

☐ Berndt-Hall-Hall-Hausman

OK Cancel

The Heteroskedasticity Consistent Covariance option is used to compute the quasi-maximum likelihood (QML) covariances and standard errors using the methods described by Bollerslev and Wooldridge (1992). This option should be used if you suspect that the residuals are not conditionally normally distributed. Note that the parameter estimates will be (virtually) unchanged if this option is selected; only the estimated covariance matrix will be altered.

The log-likelihood functions for ARCH models are often not well behaved so that convergence may not be achieved with the default estimation settings. It is possible in EViews to select the iterative algorithm (Marquardt, BHHH/Gauss Newton), to change starting values, to increase the maximum number of iterations or to adjust the convergence criteria. For example, if convergence is not achieved, or implausible parameter estimates are obtained, it is sensible to re-do the estimation using a different set of starting values and/or a different optimisation algorithm.

Once the model has been estimated, EViews provides a variety of pieces of information and procedures for inference and diagnostic checking. For example, the following options are available on the View button:

- Actual, Fitted, Residual
The residuals are displayed in various forms, such as table, graphs and standardised residuals.
- GARCH graph
This graph plots the one-step ahead standard deviation, σ_t , or the conditional variance, σ_t^2 for each observation in the sample.
- Covariance Matrix
- Coefficient Tests
- Residual Tests/Correlogram-Q statistics
- Residual Tests/Correlogram Squared Residuals
- Residual Tests/Histogram-Normality Test
- Residual Tests/ARCH LM Test.

ARCH model procedures

These options are all available by pressing the 'Proc' button following the estimation of a GARCH-type model:

- Make Residual Series
- Make GARCH Variance Series
- Forecast.

Estimating the GARCH(1,1) model for the yen-dollar ('rjpy') series using the instructions as listed above, and the default settings elsewhere would yield the results:

Dependent Variable: RJPY
 Method: ML - ARCH (Marquardt) - Normal distribution
 Date: 09/06/07 Time: 18:02
 Sample (adjusted): 7/08/2002 7/07/2007
 Included observations: 1826 after adjustments
 Convergence achieved after 10 iterations
 Presample variance: backcast (parameter = 0.7)
 $GARCH = C(2) + C(3)*RESID(-1)^2 + C(4)*GARCH(-1)$

	Coefficient	Std. Error	z-Statistic	Prob.
C	0.005518	0.009396	0.587333	0.5570
Variance Equation				
C	0.001345	0.000526	2.558748	0.0105
RESID(-1)^2	0.028436	0.004108	6.922465	0.0000
GARCH(-1)	0.964139	0.005528	174.3976	0.0000
R-squared	-0.000091	Mean dependent var		0.001328
Adjusted R-squared	-0.001738	S.D. dependent var		0.439632
S.E. of regression	0.440014	Akaike info criterion		1.139389
Sum squared resid	352.7611	Schwarz criterion		1.151459
Log likelihood	-1036.262	Hannan-Quinn criter.		1.143841
Durbin-Watson stat	1.981759			

The coefficients on both the lagged squared residual and lagged conditional variance terms in the conditional variance equation are highly statistically significant. Also, as is typical of GARCH model estimates for financial asset returns data, the sum of the coefficients on the lagged squared error and lagged conditional variance is very close to unity (approximately 0.99). This implies that shocks to the conditional variance will be highly persistent. This can be seen by considering the equations for forecasting future values of the conditional variance using a GARCH model given in a subsequent section. A large sum of these coefficients will imply that a large positive or a large negative return will lead future forecasts of the variance to be high for a protracted period. The individual conditional variance coefficients are also as one would expect. The variance intercept term 'C' is very small, and the 'ARCH parameter' is around

0.03 while the coefficient on the lagged conditional variance ('GARCH') is larger at 0.96.

8.10 Extensions to the basic GARCH model

Since the GARCH model was developed, a huge number of extensions and variants have been proposed. A couple of the most important examples will be highlighted here. Interested readers who wish to investigate further are directed to a comprehensive survey by Bollerslev *et al.* (1992).

Many of the extensions to the GARCH model have been suggested as a consequence of perceived problems with standard $\text{GARCH}(p, q)$ models. First, the non-negativity conditions may be violated by the estimated model. The only way to avoid this for sure would be to place artificial constraints on the model coefficients in order to force them to be non-negative. Second, GARCH models cannot account for leverage effects (explained below), although they can account for volatility clustering and leptokurtosis in a series. Finally, the model does not allow for any direct feedback between the conditional variance and the conditional mean.

Some of the most widely used and influential modifications to the model will now be examined. These may remove some of the restrictions or limitations of the basic model.

8.11 Asymmetric GARCH models

One of the primary restrictions of GARCH models is that they enforce a symmetric response of volatility to positive and negative shocks. This arises since the conditional variance in equations such as (8.39) is a function of the magnitudes of the lagged residuals and not their signs (in other words, by squaring the lagged error in (8.39), the sign is lost). However, it has been argued that a negative shock to financial time series is likely to cause volatility to rise by more than a positive shock of the same magnitude. In the case of equity returns, such asymmetries are typically attributed to *leverage effects*, whereby a fall in the value of a firm's stock causes the firm's debt to equity ratio to rise. This leads shareholders, who bear the residual risk of the firm, to perceive their future cashflow stream as being relatively more risky.

An alternative view is provided by the 'volatility-feedback' hypothesis. Assuming constant dividends, if expected returns increase when stock

price volatility increases, then stock prices should fall when volatility rises. Although asymmetries in returns series other than equities cannot be attributed to changing leverage, there is equally no reason to suppose that such asymmetries only exist in equity returns.

Two popular asymmetric formulations are explained below: the GJR model, named after the authors Glosten, Jagannathan and Runkle (1993), and the exponential GARCH (EGARCH) model proposed by Nelson (1991).

8.12 The GJR model

The GJR model is a simple extension of GARCH with an additional term added to account for possible asymmetries. The conditional variance is now given by

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta \sigma_{t-1}^2 + \gamma u_{t-1}^2 I_{t-1} \quad (8.48)$$

where $I_{t-1} = 1$ if $u_{t-1} < 0$
 $= 0$ otherwise

For a leverage effect, we would see $\gamma > 0$. Notice now that the condition for non-negativity will be $\alpha_0 > 0$, $\alpha_1 > 0$, $\beta \geq 0$, and $\alpha_1 + \gamma \geq 0$. That is, the model is still admissible, even if $\gamma < 0$, provided that $\alpha_1 + \gamma \geq 0$.

Example 8.1

To offer an illustration of the GJR approach, using monthly S&P500 returns from December 1979 until June 1998, the following results would be obtained, with t -ratios in parentheses

$$y_t = 0.172 \quad (3.198) \quad (8.49)$$

$$\sigma_t^2 = 1.243 + 0.015u_{t-1}^2 + 0.498\sigma_{t-1}^2 + 0.604u_{t-1}^2 I_{t-1} \quad (16.372) \quad (0.437) \quad (14.999) \quad (5.772) \quad (8.50)$$

Note that the asymmetry term, γ , has the correct sign and is significant. To see how volatility rises more after a large negative shock than a large positive one, suppose that $\sigma_{t-1}^2 = 0.823$, and consider $\hat{u}_{t-1} = \pm 0.5$. If $\hat{u}_{t-1} = 0.5$, this implies that $\sigma_t^2 = 1.65$. However, a shock of the same magnitude but of opposite sign, $\hat{u}_{t-1} = -0.5$, implies that the fitted conditional variance for time t will be $\sigma_t^2 = 1.80$.

8.13 The EGARCH model

The exponential GARCH model was proposed by Nelson (1991). There are various ways to express the conditional variance equation, but one possible specification is given by

$$\ln(\sigma_t^2) = \omega + \beta \ln(\sigma_{t-1}^2) + \gamma \frac{u_{t-1}}{\sqrt{\sigma_{t-1}^2}} + \alpha \left[\frac{|u_{t-1}|}{\sqrt{\sigma_{t-1}^2}} - \sqrt{\frac{2}{\pi}} \right] \quad (8.51)$$

The model has several advantages over the pure GARCH specification. First, since the $\ln(\sigma_t^2)$ is modelled, then even if the parameters are negative, σ_t^2 will be positive. There is thus no need to artificially impose non-negativity constraints on the model parameters. Second, asymmetries are allowed for under the EGARCH formulation, since if the relationship between volatility and returns is negative, γ , will be negative.

Note that in the original formulation, Nelson assumed a Generalised Error Distribution (GED) structure for the errors. GED is a very broad family of distributions that can be used for many types of series. However, owing to its computational ease and intuitive interpretation, almost all applications of EGARCH employ conditionally normal errors as discussed above rather than using GED.

8.14 GJR and EGARCH in EViews

The main menu screen for GARCH estimation demonstrates that a number of variants on the standard GARCH model are available. Arguably most important of these are asymmetric models, such as the TGARCH ('threshold' GARCH), which is also known as the GJR model, and the EGARCH model. To estimate a GJR model in EViews, from the GARCH model equation specification screen (screenshot 8.1 above), change the 'Threshold Order' number from 0 to 1. To estimate an EGARCH model, change the 'GARCH/TARCH' model estimation default to 'EGARCH'.

Coefficient estimates for each of these specifications using the daily Japanese yen-US dollar returns data are given in the next two output tables, respectively. For both specifications, the asymmetry terms ('(RESID<0)* ARCH(1)' in the GJR model and 'RESID(-1)/@SQRT(GARCH(-1))') are not statistically significant (although it is almost significant in the case of the EGARCH model). Also in both cases, the coefficient estimates are negative, suggesting that positive shocks imply a higher next period conditional variance than negative shocks of the same sign.

Dependent Variable: RJPY
 Method: ML – ARCH (Marquardt) – Normal distribution
 Date: 09/06/07 Time: 18:20
 Sample (adjusted): 7/08/2002 7/07/2007
 Included observations: 1826 after adjustments
 Convergence achieved after 9 iterations
 Presample variance: backcast (parameter = 0.7)

$$\text{GARCH} = C(2) + C(3)*\text{RESID}(-1)^2 + C(4)*\text{RESID}(-1)^2*(\text{RESID}(-1)<0) + C(5)*\text{GARCH}(-1)$$

	Coefficient	Std. Error	z-Statistic	Prob.
C	0.005588	0.009602	0.581934	0.5606
Variance Equation				
C	0.001361	0.000544	2.503534	0.0123
$\text{RESID}(-1)^2$	0.029036	0.005373	5.404209	0.0000
$\text{RESID}(-1)^2*(\text{RESID}(-1)<0)$	-0.001027	0.006140	-0.167301	0.8671
$\text{GARCH}(-1)$	0.963989	0.005644	170.7852	0.0000
R-squared	-0.000094	Mean dependent var		0.001328
Adjusted R-squared	-0.002291	S.D. dependent var		0.439632
S.E. of regression	0.440135	Akaike info criterion		1.140477
Sum squared resid	352.7622	Schwarz criterion		1.155564
Log likelihood	-1036.256	Hannan-Quinn criter.		1.146042
Durbin-Watson stat	1.981753			

Dependent Variable: RJPY
 Method: ML – ARCH (Marquardt) – Normal distribution
 Date: 09/06/07 Time: 18:18
 Sample (adjusted): 7/08/2002 7/07/2007
 Included observations: 1826 after adjustments
 Convergence achieved after 12 iterations
 Presample variance: backcast (parameter = 0.7)

$$\text{LOG}(\text{GARCH}) = C(2) + C(3)*\text{ABS}(\text{RESID}(-1)/\text{SQRT}(\text{GARCH}(-1))) + C(4)*\text{RESID}(-1)/\text{SQRT}(\text{GARCH}(-1)) + C(5)*\text{LOG}(\text{GARCH}(-1))$$

	Coefficient	Std. Error	z-Statistic	Prob.
C	0.003756	0.010025	0.374722	0.7079
Variance Equation				
C(2)	-1.262782	0.194243	-6.501047	0.0000
C(3)	0.214215	0.034226	6.258919	0.0000
C(4)	-0.046461	0.024983	-1.859751	0.0629
C(5)	0.329164	0.112572	2.924037	0.0035
R-squared	-0.000031	Mean dependent var		0.001328
Adjusted R-squared	-0.002227	S.D. dependent var		0.439632
S.E. of regression	0.440121	Akaike info criterion		1.183216
Sum squared resid	352.7398	Schwarz criterion		1.198303
Log likelihood	-1075.276	Hannan-Quinn criter.		1.188781
Durbin-Watson stat	1.981879			

This is the opposite to what would have been expected in the case of the application of a GARCH model to a set of stock returns. But arguably, neither the *leverage effect* or *volatility feedback* explanations for asymmetries in the context of stocks apply here. For a positive return shock, this implies more yen per dollar and therefore a strengthening dollar and a weakening yen. Thus the results suggest that a strengthening dollar (weakening yen) leads to higher next period volatility than when the yen strengthens by the same amount.

8.15 Tests for asymmetries in volatility

Engle and Ng (1993) have proposed a set of tests for asymmetry in volatility, known as sign and size bias tests. The Engle and Ng tests should thus be used to determine whether an asymmetric model is required for a given series, or whether the symmetric GARCH model can be deemed adequate. In practice, the Engle–Ng tests are usually applied to the residuals of a GARCH fit to the returns data. Define S_{t-1}^- as an indicator dummy that takes the value 1 if $\hat{u}_{t-1} < 0$ and zero otherwise. The test for sign bias is based on the significance or otherwise of ϕ_1 in

$$\hat{u}_t^2 = \phi_0 + \phi_1 S_{t-1}^- + v_t \quad (8.52)$$

where v_t is an iid error term. If positive and negative shocks to $\hat{u}_{t-1}$ impact differently upon the conditional variance, then ϕ_1 will be statistically significant.

It could also be the case that the magnitude or size of the shock will affect whether the response of volatility to shocks is symmetric or not. In this case, a negative size bias test would be conducted, based on a regression where S_{t-1}^- is now used as a slope dummy variable. Negative size bias is argued to be present if ϕ_1 is statistically significant in the regression

$$\hat{u}_t^2 = \phi_0 + \phi_1 S_{t-1}^- u_{t-1} + v_t \quad (8.53)$$

Finally, defining $S_{t-1}^+ = 1 - S_{t-1}^-$, so that S_{t-1}^+ picks out the observations with positive innovations, Engle and Ng propose a joint test for sign and size bias based on the regression

$$\hat{u}_t^2 = \phi_0 + \phi_1 S_{t-1}^- + \phi_2 S_{t-1}^- u_{t-1} + \phi_3 S_{t-1}^+ u_{t-1} + v_t \quad (8.54)$$

Significance of ϕ_1 indicates the presence of sign bias, where positive and negative shocks have differing impacts upon future volatility, compared with the symmetric response required by the standard GARCH

formulation. On the other hand, the significance of ϕ_2 or ϕ_3 would suggest the presence of size bias, where not only the sign but the magnitude of the shock is important. A joint test statistic is formulated in the standard fashion by calculating TR^2 from regression (8.54), which will asymptotically follow a χ^2 distribution with 3 degrees of freedom under the null hypothesis of no asymmetric effects.

8.15.1 News impact curves

A pictorial representation of the degree of asymmetry of volatility to positive and negative shocks is given by the news impact curve introduced by Pagan and Schwert (1990). The news impact curve plots the next-period volatility (σ_t^2) that would arise from various positive and negative values of u_{t-1} , given an estimated model. The curves are drawn by using the estimated conditional variance equation for the model under consideration, with its given coefficient estimates, and with the lagged conditional variance set to the unconditional variance. Then, successive values of u_{t-1} are used in the equation to determine what the corresponding values of σ_t^2 derived from the model would be. For example, consider the GARCH and GJR model estimates given above for the S&P500 data from EViews. Values of u_{t-1} in the range $(-1, +1)$ are substituted into the equations in each case to investigate the impact on the conditional variance during the next period. The resulting news impact curves for the GARCH and GJR models are given in figure 8.3.

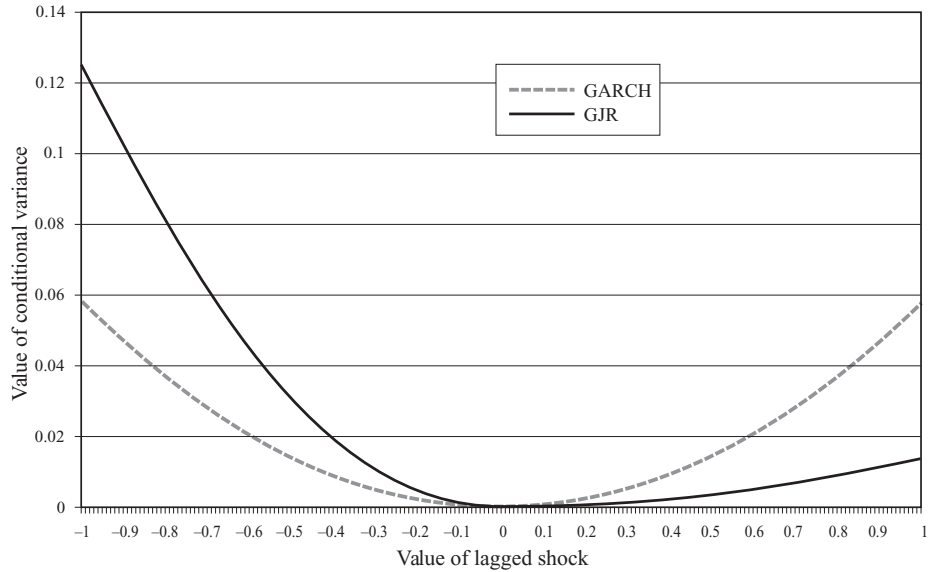
As can be seen from figure 8.3, the GARCH news impact curve (the grey line) is of course symmetrical about zero, so that a shock of given magnitude will have the same impact on the future conditional variance whatever its sign. On the other hand, the GJR news impact curve (the black line) is asymmetric, with negative shocks having more impact on future volatility than positive shocks of the same magnitude. It can also be seen that a negative shock of given magnitude will have a bigger impact under GJR than would be implied by a GARCH model, while a positive shock of given magnitude will have more impact under GARCH than GJR. The latter result arises as a result of the reduction in the value of α_1 , the coefficient on the lagged squared error, when the asymmetry term is included in the model.

8.16 GARCH-in-mean

Most models used in finance suppose that investors should be rewarded for taking additional risk by obtaining a higher return. One way to

Figure 8.3

News impact curves for S&P500 returns using coefficients implied from GARCH and GJR model estimates



operationalise this concept is to let the return of a security be partly determined by its risk. Engle, Lilien and Robins (1987) suggested an ARCH-M specification, where the conditional variance of asset returns enters into the conditional mean equation. Since GARCH models are now considerably more popular than ARCH, it is more common to estimate a GARCH-M model. An example of a GARCH-M model is given by the specification

$$y_t = \mu + \delta\sigma_{t-1} + u_t, u_t \sim N(0, \sigma_t^2) \quad (8.55)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta\sigma_{t-1}^2 \quad (8.56)$$

If δ is positive and statistically significant, then increased risk, given by an increase in the conditional variance, leads to a rise in the mean return. Thus δ can be interpreted as a risk premium. In some empirical applications, the conditional variance term, σ_{t-1}^2 , appears directly in the conditional mean equation, rather than in square root form, σ_{t-1} . Also, in some applications the term is contemporaneous, σ_t^2 , rather than lagged.

8.16.1 GARCH-M estimation in EViews

The GARCH-M model with the conditional standard deviation term in the mean, estimated using the rjpy data in EViews from the main GARCH menu as described above, would give the following results:

Dependent Variable: RJPY

Method: ML – ARCH (Marquardt) – Normal distribution

Date: 09/06/07 Time: 18:58

Sample (adjusted): 7/08/2002 7/07/2007

Included observations: 1826 after adjustments

Convergence achieved after 18 iterations

Presample variance: backcast (parameter = 0.7)

GARCH = C(3) + C(4)*RESID(−1)^2 + C(5)*GARCH(−1)

	Coefficient	Std. Error	z-Statistic	Prob.
SQRT(GARCH)	−0.068943	0.124958	−0.551729	0.5811
C	0.033279	0.051802	0.642436	0.5206
Variance Equation				
C	0.001373	0.000529	2.594929	0.0095
RESID(−1)^2	0.028886	0.004150	6.960374	0.0000
GARCH(−1)	0.963568	0.005580	172.6828	0.0000
R-squared	0.000034	Mean dependent var		0.001328
Adjusted R-squared	−0.002162	S.D. dependent var		0.439632
S.E. of regression	0.440107	Akaike info criterion		1.140302
Sum squared resid	352.7170	Schwarz criterion		1.155390
Log likelihood	−1036.096	Hannan-Quinn criter.		1.145867
F-statistic	0.015541	Durbin-Watson stat		1.982106
Prob(F-statistic)	0.999526			

In this case, the estimated parameter on the mean equation has a negative sign but is not statistically significant. We would thus conclude that for these currency returns, there is no feedback from the conditional variance to the conditional mean.

8.17 Uses of GARCH-type models including volatility forecasting

Essentially GARCH models are useful because they can be used to model the volatility of a series over time. It is possible to combine together more than one of the time series models that have been considered so far in this book, to obtain more complex ‘hybrid’ models. Such models can account for a number of important features of financial series at the same time – e.g. an ARMA-EGARCH(1,1)-M model; the potential complexity of the model is limited only by the imagination!

GARCH-type models can be used to forecast volatility. GARCH is a model to describe movements in the conditional variance of an error term,

u_t , which may not appear particularly useful. But it is possible to show that

$$\text{var}(y_t | y_{t-1}, y_{t-2}, \dots) = \text{var}(u_t | u_{t-1}, u_{t-2}, \dots) \quad (8.57)$$

So the conditional variance of y , given its previous values, is the same as the conditional variance of u , given its previous values. Hence, modelling σ_t^2 will give models and forecasts for the variance of y_t as well. Thus, if the dependent variable in a regression, y_t is an asset return series, forecasts of σ_t^2 will be forecasts of the future variance of y_t . So one primary usage of GARCH-type models is in forecasting volatility. This can be useful in, for example, the pricing of financial options where volatility is an input to the pricing model. For example, the value of a ‘plain vanilla’ call option is a function of the current value of the underlying, the strike price, the time to maturity, the risk free interest rate and volatility. The required volatility, to obtain an appropriate options price, is really the volatility of the underlying asset expected over the lifetime of the option. As stated previously, it is possible to use a simple historical average measure as the forecast of future volatility, but another method that seems more appropriate would be to use a time series model such as GARCH to compute the volatility forecasts. The forecasting ability of various models is considered in a paper by Day and Lewis (1992), discussed in detail below.

Producing forecasts from models of the GARCH class is relatively simple, and the algebra involved is very similar to that required to obtain forecasts from ARMA models. An illustration is given by example 8.2.

Example 8.2

Consider the following GARCH(1,1) model

$$y_t = \mu + u_t, u_t \sim N(0, \sigma_t^2) \quad (8.58)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (8.59)$$

Suppose that the researcher had estimated the above GARCH model for a series of returns on a stock index and obtained the following parameter estimates: $\hat{\mu} = 0.0023$, $\hat{\alpha}_0 = 0.0172$, $\hat{\beta} = 0.7811$, $\hat{\alpha}_1 = 0.1251$. If the researcher has data available up to and including time T , write down a set of equations in σ_t^2 and u_t^2 and their lagged values, which could be employed to produce one-, two-, and three-step-ahead forecasts for the conditional variance of y_t .

What is needed is to generate forecasts of $\sigma_{T+1}^2 | \Omega_T$, $\sigma_{T+2}^2 | \Omega_T$, $\dots$, $\sigma_{T+s}^2 | \Omega_T$ where Ω_T denotes all information available up to and including

observation T . For time T , the conditional variance equation is given by (8.59). Adding one to each of the time subscripts of this equation, and then two, and then three would yield equations (8.60)–(8.62)

$$\sigma_{T+1}^2 = \alpha_0 + \alpha_1 u_T^2 + \beta \sigma_T^2 \quad (8.60)$$

$$\sigma_{T+2}^2 = \alpha_0 + \alpha_1 u_{T+1}^2 + \beta \sigma_{T+1}^2 \quad (8.61)$$

$$\sigma_{T+3}^2 = \alpha_0 + \alpha_1 u_{T+2}^2 + \beta \sigma_{T+2}^2 \quad (8.62)$$

Let $\sigma_{1,T}^{f^2}$ be the one-step-ahead forecast for σ^2 made at time T . This is easy to calculate since, at time T , the values of all the terms on the RHS are known. $\sigma_{1,T}^{f^2}$ would be obtained by taking the conditional expectation of (8.60).

Given $\sigma_{1,T}^{f^2}$, how is $\sigma_{2,T}^{f^2}$, the two-step-ahead forecast for σ^2 made at time T , calculated?

$$\sigma_{1,T}^{f^2} = \alpha_0 + \alpha_1 u_T^2 + \beta \sigma_T^2 \quad (8.63)$$

From (8.61), it is possible to write

$$\sigma_{2,T}^{f^2} = \alpha_0 + \alpha_1 E(u_{T+1}^2 | \Omega_T) + \beta \sigma_{1,T}^{f^2} \quad (8.64)$$

where $E(u_{T+1}^2 | \Omega_T)$ is the expectation, made at time T , of u_{T+1}^2 , which is the squared disturbance term. It is necessary to find $E(u_{T+1}^2 | \Omega_T)$, using the expression for the variance of a random variable u_t . The model assumes that the series u_t has zero mean, so that the variance can be written

$$\text{var}(u_t) = E[(u_t - E(u_t))^2] = E(u_t^2). \quad (8.65)$$

The conditional variance of u_t is σ_t^2 , so

$$\sigma_t^2 | \Omega_t = E(u_t^2) \quad (8.66)$$

Turning this argument around, and applying it to the problem at hand

$$E(u_{T+1}^2 | \Omega_t) = \sigma_{T+1}^2 \quad (8.67)$$

but σ_{T+1}^2 is not known at time T , so it is replaced with the forecast for it, $\sigma_{1,T}^{f^2}$, so that (8.64) becomes

$$\sigma_{2,T}^{f^2} = \alpha_0 + \alpha_1 \sigma_{1,T}^{f^2} + \beta \sigma_{1,T}^{f^2} \quad (8.68)$$

$$\sigma_{2,T}^{f^2} = \alpha_0 + (\alpha_1 + \beta) \sigma_{1,T}^{f^2} \quad (8.69)$$

What about the three-step-ahead forecast?

By similar arguments,

$$\sigma_{3,T}^{f^2} = E_T(\alpha_0 + \alpha_1 u_{T+2}^2 + \beta \sigma_{T+2}^2) \quad (8.70)$$

$$\sigma_{3,T}^{f^2} = \alpha_0 + (\alpha_1 + \beta) \sigma_{2,T}^{f^2} \quad (8.71)$$

$$\sigma_{3,T}^{f^2} = \alpha_0 + (\alpha_1 + \beta) [\alpha_0 + (\alpha_1 + \beta) \sigma_{1,T}^{f^2}] \quad (8.72)$$

$$\sigma_{3,T}^{f^2} = \alpha_0 + \alpha_0(\alpha_1 + \beta) + (\alpha_1 + \beta)^2 \sigma_{1,T}^{f^2} \quad (8.73)$$

Any s -step-ahead forecasts would be produced by

$$\sigma_{s,T}^{f^2} = \alpha_0 \sum_{i=1}^{s-1} (\alpha_1 + \beta)^{i-1} + (\alpha_1 + \beta)^{s-1} \sigma_{1,T}^{f^2} \quad (8.74)$$

for any value of $s \geq 2$.

It is worth noting at this point that variances, and therefore variance forecasts, are additive over time. This is a very useful property. Suppose, for example, that using daily foreign exchange returns, one-, two-, three-, four-, and five-step-ahead variance forecasts have been produced, i.e. a forecast has been constructed for each day of the next trading week. The forecasted variance for the whole week would simply be the sum of the five daily variance forecasts. If the standard deviation is the required volatility estimate rather than the variance, simply take the square root of the variance forecasts. Note also, however, that standard deviations are not additive. Hence, if daily standard deviations are the required volatility measure, they must be squared to turn them to variances. Then the variances would be added and the square root taken to obtain a weekly standard deviation.

8.17.1 Forecasting from GARCH models with EViews

Forecasts from any of the GARCH models that can be estimated using EViews are obtained by using only a sub-sample of available data for model estimation, and then by clicking on the 'Forecast' button that appears after the estimation of the required model has been completed. Suppose, for example, we stopped the estimation of the GARCH(1,1) model for the Japanese yen returns on 6 July 2005 so as to keep the last two years of data for forecasting (i.e. the 'Forecast sample' is **7/07/2005 7/07/2007**). Then click **Proc/Forecast...** and the dialog box in screenshot 8.3 will then appear.

Again, several options are available, including providing a name for the conditional mean and for the conditional variance forecasts, or whether to produce static (a series of rolling single-step-ahead) or dynamic (multiple-step-ahead) forecasts. The dynamic and static forecast plots that would be produced are given in screenshots 8.4 and 8.5.